Austerity, Democracy, and Coercion: A Model of Fiscal Pressure and the Severity of Protest Policing

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Abstract

Democratic governments decide in advance how harshly police may treat protesters, and preparing for a large crowd is expensive. Much is known about whether repression deters or provokes protest, less about how the cost of managing a crowd affects authorized harm. This paper models an elected executive who sets a policing directive before citizens decide whether to join a protest that succeeds only if enough participate. A harsher directive discourages participation and lets the state prepare for a smaller crowd, but raises legal, electoral, and operational costs. When preparing for each protester becomes costlier, the state authorizes more harm if the resources saved by deterring participants exceed the added cost of harsher tactics. Budget pressure therefore produces harsher repression where policing costs lie in deployment, and restraint where the same inputs implement force. Stronger accountability lowers expected harm, as does anticipated protester anger where policing costs lie in deployment.

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1 Introduction.

Mass protests are costly for incumbent elites in democracies for two reasons. A massive protest may force politicians to change policies they prefer to preserve (Amenta et al. 2010; Klein and Regan 2018). Massive protests also require the state to deploy personnel, equipment, transport, detention capacity, and other operational resources, and these resources must be planned before turnout is known. Elected executives and police commanders set deployments, authorized tactics, and rules of engagement for anticipated protest episodes, within the framework set by legal rules, standing procedures, and political oversight (della Porta and Reiter 1998, pp. 9, 15–17). A harsher directive can deter participation and thus reduce the crowd the state must prepare for. This paper asks whether the cost of managing a crowd affects how severely a democratic state polices protest. Specifically, we ask whether a government authorizes more harm against protesters, or less, when preparing for them becomes more expensive.

Officers and commanders already weigh this tradeoff. Greek riot police interviewed by Vasilaki (2016, p. 20) name numerical shortfall directly: "Operations are designed in a way that makes us violent. We are few against many." British officers interviewed after the 2011 riots attributed their loss of control to a lack of resources, and the Metropolitan Police afterward sought water cannon and authorized baton rounds (O'Sullivan 2017, pp. 35–36). Yet the same budget pressure also produced restraint, since the Metropolitan Police extended protest liaison partly because it could reduce the police resources mobilized at protests (p. 198). These accounts show that the question arises in practice, but leave open whether budget pressure raises or lowers the harm a government authorizes.

Studies disagree about how state repression affects mobilization (Davenport 2007a). Some find that threats of repression deter mobilization by raising the cost of protest (Chenoweth, Perkoski, and Kang 2017; della Porta 1995; C. M. Sullivan 2016). Others find that repression can backfire by radicalizing victims of state violence (Adam-Troian, Çelebi, and Mahfud 2020; C. M. Sullivan and Davenport 2017). Scholars have studied

this nexus between protest and repression primarily in autocracies (Davenport 2007a; Curtice and Behlendorf 2021). An empirical strand examines protest in democracies (Carey 2006; Nam 2007; Machado, Scartascini, and Tommasi 2011). It asks mainly why protest occurs more often in some democratic settings than in others (Su 2015; Aytaç, Schiumerini, and Stokes 2017). Competitive elections, protected speech, and organized civil society make protest a viable complement to representative channels, while legal and electoral accountability make the costs of coercion especially visible. Democratic protest policing therefore places deterrence, operational capacity, and political accountability in the same strategic problem.

Neighboring literatures have not theorized how the cost of managing a crowd shapes the severity of authorized protest policing. Research on protest policing explains the choice of tactics mainly through threat perception, security diffusion, and legal change (Noakes and Gillham 2007; Gillham 2011; Selmini and Di Ronco 2023). None of these accounts examines how the structure of police costs shapes the choice of tactics. Studies of capacity and fiscal pressure come closer. H. Sullivan (2021) finds that Mexican subnational governments with bureaucratic capacity respond to protest with repression less often. Beck (2023) finds that United States municipalities that rely more on fines and fees experience more police killings, a fiscal determinant of coercive severity outside the protest setting. Bueno de Mesquita and Shadmehr (2023) provide the closest application of global games to repression, and their government pays a single cost of its repression level. That cost does not vary with the size of the crowd, and the model has no separate political cost of authorizing harm.

To study this question, we develop a sequential model in which citizens protest to change a policy. The elected executive first sets the expected repression cost borne by a participant. This measure combines the reach and intensity of coercion. A continuum of citizens then decides whether to protest. The protest changes policy only if participation reaches an uncertain critical threshold, and each citizen observes a noisy private signal

of that threshold. Participants receive an expressive reward only if the protest succeeds, and they pay the expected repression cost. The state values preservation of the disputed status quo. Citizens' private information concerns only the conditions for success, not the merits of the policy, so protest changes policy through pressure, not by informing the government as in Battaglini (2017). The state plans operational resources for the expected volume of protest and pays a convex political cost from expected participant harm. We take the participation stage from the psychological rewards model of Bueno de Mesquita and Shadmehr (2023) and give the state a payoff that separates these two costs.

The central result is a condition on how the police organization produces force. When preparing for each additional protester becomes more expensive, the state authorizes more expected harm if the resources saved by deterring participants exceed the added cost of implementing harsher tactics. The logic has two steps. Expected harm discourages some citizens from joining, and a smaller expected crowd lets the government plan fewer officers, vehicles, detention spaces, and supplies. Thus, budget pressure produces harsher repression where policing costs lie mainly in deployment. Yet the same budget pressure can instead produce restraint if stronger tactics require specialized officers, equipment, documentation, medical support, or detention capacity. What matters is the cost of managing one more protester. The total price of maintaining a police force is a different quantity. Additionally, stronger accountability lowers expected harm. An extension shows that repression backfires only when protesters' motivation rises faster than the harm they expect, and that a state anticipating a stronger emotional response authorizes less harm when its costs lie in deployment.

The model also points empirical work toward specific measurements. First, expected participant harm combines reach, the probability that a participant is targeted, with the harm conditional on targeting, and event records can code both. Second, researchers can distinguish standing capacity from resources added in anticipation of protest, using budget documents, deployment plans, procurement records, and reports written after each

episode. The same records can show whether a cost change falls mainly on deployment or on implementing force, which determines the sign of the budget effect. Third, because the directive is a written policy, changes in authorized force can be dated in rules of engagement, internal directives, and less-lethal procurement, as in the Zurich case discussed below (Wisler and Kriesi 1998, p. 100), and compared with the timing of personnel and budget shocks.

The rest of the paper situates the argument in research on repression, mobilization, and coordination, presents the participation stage, develops the state's problem and its comparative statics, and discusses the paper's contribution and its empirical implications.

2 Repression, mobilization, and protest coordination.

2.1 State repression and protest mobilization.

Research on state repression examines why governments coerce citizens and how coercion affects political behavior, especially protest. Political repression is the use of institutional or physical force, most commonly violent, to subdue or coerce individuals and groups that challenge political actors or the status quo (deMeritt 2016). Much of this research focuses on authoritarian regimes or flawed democracies, where rulers face weak accountability and citizens lack protected speech or competitive elections (della Porta and Reiter 1998, p. 1).

Comparative research has persistently found that democratic countries exhibit lower levels of state repression (Davenport 2007a, 2007b; Davenport and Armstrong 2004). Yet violent coercion by state agents also occurs in developed democracies (Arias and Goldstein 2010; Belur 2010; Worden 1995). Citizen backlash is not confined to force used against protesters. George Floyd was killed during an arrest, not while participating in a demonstration. The ensuing protests reduced favorability toward the police among low-prejudice and politically liberal Americans (Reny and Newman 2021). Police violence

outside a protest can therefore mobilize citizens and raise the political costs that surround later police action. Victims of police repression may also experience anxiety, depression, anger, and fear (Bor et al. 2018; Çelebi, Adam-Troian, and Mahfud 2022; Cooper et al. 2004; J. E. DeVylder et al. 2017; J. DeVylder, Fedina, and Link 2020).

Davenport (2007a) calls the conflicting evidence about how repression affects social movements the "punishment puzzle." Some studies find that repression discourages mass and peaceful protest but encourages radicalization (della Porta 1995). Others find that it increases both protest escalation and mobilization (Adam-Troian, Çelebi, and Mahfud 2020; C. M. Sullivan and Davenport 2017).

2.2 The strategic interdependence of dissent and repression.

The literature has identified several determinants that systematically explain state repression, among them regime type, economic development, population size, intrastate civil conflicts, and the degree of grievances and dissent (Davenport 2007b; Nordås and Davenport 2013; C. M. Sullivan 2016; C. M. Sullivan and Davenport 2017). For example, Nordås and Davenport (2013) argue that the demographic structure of the population is a key determinant of state repression. States respond with more repression where youth bulges are larger, meaning cohorts aged 15 to 24 relative to the overall adult population. States that anticipate more dissent raise repression preemptively, to deter it.

Chenoweth, Perkoski, and Kang (2017) summarize several consistent findings in the literature. First, dissent precedes state repression in most settings, a regularity Davenport (2007a, p. 7) calls the "law of coercive responsiveness." Second, democratic countries tend to exhibit lower levels of repression. The authors also discuss the "punishment puzzle" mentioned above. They stress that repression and dissent, whether protest or other tactics, are two complex interlinked processes.

Formal models explain the relationship between state repression, protests, and regime or policy change in several ways. Ritter (2014) notes that models choose different objects

of investigation, and that each one, taken alone, misses important strategic features of the relation between repression and dissent. For example, some studies model the determinants of regime change or dissent in general as a process of an ongoing or expected conflict. In one approach, citizens protest to alter the status quo, and their success depends on the state's concessions. A revolutionary vanguard's level of violence may signal the level of dissent in society, which provokes citizens to cooperate (Bueno de Mesquita 2010). Other formal models turn to uncertainty and asymmetric information about government types (Pierskalla 2010). A "strong" state may repress citizens at high cost to signal its resolve. Citizens protest over repeated play and learn what type of government they face. Closest to the present paper in timing, De Jaegher and Hoyer (2019) model *preemptive* repression. A government sets repressive capacity before protest occurs, and dissidents respond by investing in countermeasures. De Jaegher and Hoyer derive both sides of the punishment puzzle. Repression deters when the benefit of protest is large relative to the cost of countermeasures, and backfires when it is small. Their complete-information coordination game has two dissidents, each of whom can determine the outcome. The relative size of the equilibria's basins of attraction resolves the resulting multiplicity.

Thus, as Ritter (2014) notes, some models study the conditions that ignite dissent, and others study how the state chooses the severity of repression. But studies miss an important part of the puzzle when they treat these two processes separately. She therefore proposes a model that considers the conditions sparking dissent together with the state's choice of severity. She models the strategic setting as a bargaining problem over policy provision and its effect on incumbent survival. Ritter finds that a higher likelihood of leader survival decreases the level of dissent and repression. Yet the same increase raises the severity of both.

2.3 Global games of protest and regime change.

This paper borrows its analytic setup from the global games literature on coordination under incomplete information (Carlsson and van Damme 1993; Morris and Shin 1998, 2003). The canonical regime change game has a continuum of citizens who simultaneously decide whether to attack a regime. The regime collapses if and only if the mass of attackers exceeds its strength. That strength is an uncertain fundamental, and citizens receive noisy private signals about it. Under complete information about the fundamental, the game exhibits multiple equilibria in an intermediate range of regime strength. The incomplete information selects a unique equilibrium in threshold strategies.

Bueno de Mesquita and Shadmehr (2023) provide the closest application of this framework to repression. Their psychological-rewards model follows the standard global-game setup for regime change (p. 738). A continuum of citizens receive noisy private signals $x_i = \theta + \sigma\varepsilon_i$ about the regime's strength θ, and the rebellion succeeds if and only if the fraction of rebels exceeds θ. Their psychological-rewards setting yields the threshold $1 - c$, where c is the participation cost, and their government chooses costly repressive capacity before citizens move. The present paper takes that participation stage as its benchmark. Its contribution begins with the government's payoff. The state values a disputed policy status quo rather than regime survival, and it separately accounts for the operational resources planned for expected protest volume and the political cost of expected participant harm. This distinction produces the crowd-management mechanism studied below.

The success-contingent reward also connects the model to work on selective incentives and democratic participation. Aytaç and Stokes (2019) explain participation through costs that citizens associate with abstention, including emotional responses and social obligations. Wood's (2003) pleasure-in-agency similarly locates a reward in contributing rather than in passively receiving a public good. These approaches distinguish participation-contingent motivation from the policy benefit shared by participants and nonparticipants. The distinction supplies the strategic complementarity in the benchmark and clarifies the

conditional role of the citizens' policy prize.

Little (2016) distinguishes two coordination problems that potential protesters face. The first is "political coordination," deciding whether to participate without knowing how many others are dissatisfied enough to join. The second is "tactical coordination," deciding when, where, and how to protest. His political coordination model also has a continuum of citizens with heterogeneous assessments and equilibria in cutoff strategies. He argues that better tactical information generally makes mobilization less costly and opposition action larger and more cohesive (p. 152).

Chau, Hassan, and Little (2025) show how private communication allows protesters to coordinate across sites and stretch government forces, whereas surveillance erodes that advantage. We set aside communication and site choice to examine how the cost of preparing for a crowd affects the harm the state authorizes before turnout is known.

3 A crowd-cost model of protest policing.

In this section, we develop a model of citizens who decide to protest to change a disliked policy. The citizens face a coordination and collective action problem. The protest changes the policy only if citizens' participation reaches a given threshold. The threshold is an uncertain fundamental, and citizens receive noisy private signals of it. On the other side, the state would prefer not to change the policy. To influence participation, it can set the expected repression cost borne by a participant. A realized mass protest consumes operational resources, while a policy that imposes greater expected harm invites legal, electoral, and reputational backlash. We begin with the citizens' participation problem and its complete-information benchmark. Next, we solve the global game. Then, we propose the two-stage sequential game in which the state sets expected participant cost first, and we inspect the strategic implications of the model.

Table 1: Notation and role in the model

Object	Meaning and units	Role
τ	Critical participating share required for policy change	Realized fundamental
N^*	Share of citizens who participate	Equilibrium outcome
B_p	Public policy benefit from successful protest	Citizen payoff primitive
d_p	Success-contingent participation reward	Citizen payoff primitive
q_r	Probability or share of participants targeted	Measured component of the directive
s_r	Harm conditional on being targeted	Measured component of the directive
$c_r = q_r s_r$	Expected repression cost per participant	State choice
B_s	Government value of preserving the policy status quo	State payoff primitive
$\bar{N}(c_r)$	Expected share of citizens who participate	Anticipated outcome
$V(\bar{N}, c_r; w)$	Planned operational cost for expected protest volume and the chosen repression policy	State cost technology
$P(c_r; \gamma)$	Political cost of expected participant harm	State cost technology
w	Crowd-cost shifter: price of personnel and logistical inputs	State cost parameter
γ	Accountability: raises the marginal political price of expected participant harm	State cost parameter
σ	Noise in citizens' private signals	Information parameter

3.1 The citizens' participation problem.

There is a continuum of citizens of mass one, indexed by $i \in [0,1]$. Each citizen chooses whether to participate in the protest. Let $n_i \in \{0,1\}$ indicate participation, with $n_i = 1$ for participating and $n_i = 0$ for abstaining, and let $N^* \in [0,1]$ denote the realized mass of participants. Protesters want to change a policy so gridlocked that ordinary representative channels struggle to move it. The protest is successful if participation reaches a critical threshold τ, so the policy changes if and only if $N^* \geq \tau$.

The payoff primitives are as follows. A change in the policy provides a collective and indivisible benefit B_p to all individuals in the society (i.e., a public good). Individuals who participate may get an expressive or emotional benefit d_p. They may feel, for example, that fighting for a just cause is good in and of itself. Participants also pay c_r, the expected repression cost borne by a participant. This object combines the probability of victimization with the conditional harm suffered if victimized. If q_r denotes reach and s_r conditional severity, then $c_r = q_r s_r$.

Assumption 1 (One-dimensional directive)**.** The executive chooses from an institution-specific, one-dimensional menu of policing directives indexed by $c_r \in [0,\infty)$. The menu assigns a single policy package to each index value, with reach $q_r : [0,\infty) \to [0,1]$, conditional severity $s_r : [0,\infty) \to [0,\infty)$, and $q_r(c_r)s_r(c_r) = c_r$.

The directive is the policy unit; q_r and s_r are its measurable components. This restriction fits institutions in which rules of engagement bundle deployment, targeting, and authorized force into stable policy packages (della Porta and Reiter 1998, pp. 3–4). The menu retains the cost-minimizing package whenever an institution makes several packages with the same product feasible, and it absorbs the minimized cost into V. The functions V and P below evaluate costs along that feasible policy menu. Expected participant harm is therefore sufficient for the citizen's deterrence decision and indexes the packages compared by the executive. Conditional severity refers only to s_r.

Two features of the continuum setting must be made explicit, because they discipline how the primitives enter the participation margin.

First, each individual citizen is nonpivotal. With finitely many players, Δp_i would place the public good benefit B_p in the individual calculus. That term is the change in the probability of meeting the threshold induced by one's own participation. In the continuum, a single citizen has measure zero, so $\Delta p_i = 0$. The citizen receives B_p when the protest succeeds whether she participated or abstained, and B_p cancels from that comparison. The benchmark therefore isolates the participation-contingent reward as the motive of the marginal participant.

Second, the expressive benefit must be success-contingent for participation to respond to the prospects of success at all. A participant enjoys the expressive reward d_p only if the movement succeeds. That satisfaction arrives only when the change actually happens. This success-contingency is a substantive assumption, and it generates the coordination problem.[1] Were d_p unconditional, participation would reduce to the comparison of d_p against c_r, independent of any belief about others, and the strategic structure would disappear.

With these two features, the net payoff of participating relative to abstaining, for citizen i, is

$$u_i(1) - u_i(0) = \mathbb{1}\{N^* \geq \tau\} \cdot d_p - c_r,$$

since $B_p \cdot \mathbb{1}\{N^* \geq \tau\}$ enters both sides and cancels. A citizen therefore participates when her belief that the protest will succeed, times the expressive reward, exceeds the expected repression cost.

1. See Bueno de Mesquita and Shadmehr (2023) and Wood (2003).

3.2 The complete-information benchmark.

Suppose first, as a benchmark, that τ is common knowledge. Assume $0 < \tau \leq 1$, the interesting range, and $c_r < d_p$, so that participation is worthwhile when success is assured. Then two pure strategy equilibria coexist, and the small-noise limit selects a unique interior prediction (Carlsson and van Damme 1993; Morris and Shin 1998, 2003). If every citizen participates, $N^* = 1 \geq \tau$, the protest succeeds, and each participant receives $d_p - c_r > 0$. No citizen gains by deviating, since abstaining forfeits $d_p - c_r$ while leaving success (and hence B_p) unchanged. If no citizen participates, $N^* = 0 < \tau$, the protest fails, and a lone deviator would pay c_r for a failed protest. Universal abstention is therefore also an equilibrium. If instead $c_r > d_p$, universal abstention is the unique equilibrium, because participation is dominated even under assured success.

The complete-information benchmark presents two problems. The multiplicity makes prediction difficult, and the equilibrium set is insensitive to the parameters, including the repression policy itself. One could respond by selecting an equilibrium, for instance the mixed strategy equilibrium, and reading its participation probabilities as frequencies. We instead take seriously that the critical threshold is a variable citizens cannot observe. The size a protest must reach to force a policy change is written nowhere, unlike the threshold to win an election under plurality rule, which is common knowledge. How citizens form beliefs about τ is therefore central to any behavioral prediction.

3.3 The global game.

Let the critical threshold τ be drawn from the common prior $U[0,1]$ and remain unobserved by citizens. Citizens and the state use this same proper prior. Each citizen receives a noisy private signal

$$x_i = \tau + \sigma \varepsilon_i,$$

where the ε_i are i.i.d. with a log-concave density f and c.d.f. F with full support on $\mathbb{R}$, independent of τ, and $\sigma > 0$ scales the noise. A citizen with a low signal believes the required crowd is small and success likely. A citizen with a high signal believes success is out of reach.

A strategy is a mapping from signals to participation. The participation payoff is decreasing in the (believed) threshold, so cutoff strategies are the natural place to look for equilibria. A citizen participates if and only if $x_i < \hat{x}$. Given such a cutoff, the mass of participants at true threshold τ is

$$N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = F\left(\frac{\hat{x} - \tau}{\sigma}\right),$$

which is continuous and strictly decreasing in τ. The protest therefore succeeds if and only if τ is below a unique critical value $\hat{\tau}$. The equilibrium is characterized by two conditions:

$$\text{(belief consistency / critical mass):} \qquad F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = \hat{\tau},$$

$$\text{(indifference of the marginal citizen):} \qquad \Pr(\tau \le \hat{\tau} \mid x_i = \hat{x}) \cdot d_p = c_r.$$

The first condition says that at the critical threshold the crowd is exactly critical. The second says that a citizen whose signal equals the cutoff is indifferent between participating and abstaining. Under the proper prior, the marginal citizen's exact posterior success probability is

$$\pi_\sigma(\hat{x}, \hat{\tau}) = \frac{F(\hat{x}/\sigma) - F((\hat{x} - \hat{\tau})/\sigma)}{F(\hat{x}/\sigma) - F((\hat{x} - 1)/\sigma)}.$$

Thus a positive-noise monotone cutoff equilibrium solves

$$F\left(\frac{\hat{x}_\sigma - \hat{\tau}_\sigma}{\sigma}\right) = \hat{\tau}_\sigma, \qquad d_p\, \pi_\sigma(\hat{x}_\sigma, \hat{\tau}_\sigma) = c_r. \tag{1}$$

These equations retain the boundary terms generated by $U[0,1]$. At an interior threshold

they converge, as $\sigma \to 0$, to $(1-\hat{\tau})d_p = c_r$. The limiting success threshold is therefore

$$\hat{\tau}(c_r) \;=\; \max\left\{1 - \frac{c_r}{d_p}, 0\right\}. \tag{2}$$

Normalized expected participant harm c_r/d_p determines the limiting threshold. Proposition 1 states the proper-prior convergence result.

Proposition 1 (Proper-prior small-noise selection)**.** *The selected monotone cutoff outcome has the following properties:*

(i) Fix a policy with $\rho \equiv c_r/d_p \in (0,1)$ and let $\hat{\tau}_0 = 1-\rho$. For every $\delta \in (0, \min\{\hat{\tau}_0, 1-\hat{\tau}_0\})$, there is a $\bar{\sigma} > 0$ such that equation (1) has an interior monotone cutoff equilibrium with $\hat{\tau}_\sigma \in [\hat{\tau}_0 - \delta, \hat{\tau}_0 + \delta]$ for every $\sigma \in (0, \bar{\sigma})$.

(ii) Every selection from these local equilibria satisfies

$$\hat{x}_\sigma \longrightarrow \hat{\tau}, \qquad \hat{\tau}_\sigma \longrightarrow \hat{\tau} = 1 - \frac{c_r}{d_p}.$$

The interior monotone limiting selection is therefore unique.

(iii) If $c_r > d_p$, participation is strictly dominated. At $c_r = d_p$, full deterrence is the continuous extension of the interior small-noise selection, although other weak equilibria can coexist at equality. At $c_r = 0$, full participation is the corresponding limiting selection.

Online Appendix A.3 (p. A-4) gives the full proof, which uses the posterior lemmas of Online Appendix A.2 (p. A-2). Three remarks.

First, the selected interior branch exists for all sufficiently small positive noise and remains in a fixed compact neighborhood of $1 - c_r/d_p$. The bounded proper prior can also generate boundary-adjacent roots at positive noise. The theory follows the local branch that continues the interior global-games prediction. In the limiting game, protest succeeds if and only if $\tau \le \hat{\tau}(c_r)$, so the state's comparative statics use its unique small-noise limit.

Second, the limiting outcome changes smoothly with the repression policy. The deterrence derivative is[2]

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{1}{d_p}.$$

Each unit of expected participant cost shrinks the set of fundamentals at which the protest succeeds at a constant rate, inversely proportional to the expressive reward. Repression is blunt against a strongly motivated citizenry (high d_p), and sharp against a weakly motivated one. This is the global games rendering of a familiar intuition: intrinsic motivation erodes the deterrent effect of expected participant harm.

Third, the direct deterrence margin is governed by d_p alone. The zero partial effect of B_p is a benchmark result conditional on three assumptions: citizens form a continuum, the policy prize is public and available equally to participants and abstainers, and policy stakes are separable from success-contingent motivation. A larger policy stake may still raise d_p by intensifying grievance, identity, efficacy, or pleasure-in-agency, and it may raise the government's B_s. The model isolates the direct pivotality channel by fixing those pathways.

The state's problem is evaluated in the small-noise limit $\sigma \to 0$.[3] The cutoff $\hat{x}$ then converges to $\hat{\tau}$ and participation becomes sharp: $N^*(\tau) \to 1$ for $\tau < \hat{\tau}$ and $N^*(\tau) \to 0$ for $\tau > \hat{\tau}$. Expected protest mass is therefore $\bar{N}(c_r) \equiv \mathbb{E}_\tau[N^*(\tau) \mid c_r] = \hat{\tau}(c_r)$. The limiting environment captures the extensive margin between a mass protest and no mobilization and keeps the state's planning problem transparent.

2. The notation $\partial\hat{\tau}/\partial c_r$ denotes the change in the limiting equilibrium success threshold induced by a marginal change in expected participant cost, with d_p fixed. The expression applies on the interior region $0 < c_r < d_p$; at the clamped boundary the relevant one-sided derivative is zero.

3. The executive's optimization problem below is defined over the limiting policy mapping $c_r \mapsto \hat{\tau}(c_r)$. The positive-noise calculations in Online Appendix A.4 (p. A-6) illustrate the robustness of the selected citizen branch.

3.4 The state's problem.

The elected executive knows the policy dispute and its stakes, then fixes an operational directive before citizens receive their private information and mobilize. Police implement that directive during the episode.

Assumption 2 (Within-episode commitment)**.** The state chooses c_r after the policy dispute is known and before citizen mobilization, and c_r remains fixed during the protest episode.

The timing captures advance deployment, rules of engagement, authorized tactics, and expectations formed from directives and prior practice (della Porta and Reiter 1998, pp. 6–7). The directive is publicly announced, legally recorded, or credible under standing practice before citizens mobilize. Legal review, administrative compliance, press scrutiny, and anticipatory electoral response can therefore impose a political cost when the directive is authorized, even if deterrence prevents its use.

Let $V(\bar{N}, c_r; w)$ denote the operational cost of preparing for expected protest mass $\bar{N}$ under policy c_r. It includes personnel, equipment, transport, command capacity, and other resources committed in anticipation of protest, including preparedness expenditures incurred before turnout is realized.

Assumption 3 (Planned operational cost)**.** The function $V(\bar{N}, c_r; w)$ is twice continuously differentiable on $\bar{N} \in [0, 1]$ and $c_r \geq 0$, satisfies $V(0, 0; w) = 0$, $V_N > 0$, $V_c \geq 0$, and $V_{Nw} > 0$.

The crowd-cost shifter w raises the marginal cost of preparing for additional protesters. The cross-partial V_{cw} permits the same shifter to change the direct marginal operational cost of harsher tactics.[4]

4. Subscripts on V denote partial derivatives evaluated at $(\bar{N}, c_r; w)$. Thus $V_N \equiv \partial V/\partial \bar{N}$, $V_c \equiv \partial V/\partial c_r$, $V_{NN} \equiv \partial^2 V/\partial \bar{N}^2$, $V_{Nc} \equiv \partial^2 V/(\partial \bar{N}\partial c_r)$, $V_{cc} \equiv \partial^2 V/\partial c_r^2$, $V_{Nw} \equiv \partial^2 V/(\partial \bar{N}\partial w)$, and $V_{cw} \equiv \partial^2 V/(\partial c_r \partial w)$. Every partial derivative fixes the function's remaining arguments.

The same subscript convention applies to the political-cost function.[5]

Assumption 4 (Political cost of the authorized directive)**.** The function $P(c_r;\gamma)$ is twice continuously differentiable on $c_r \geq 0$, satisfies $P(0;\gamma) = 0$, $P_c(0;\gamma) = 0$, $P_c(c_r;\gamma) > 0$ for $c_r > 0$, $P_{cc}(c_r;\gamma) > 0$, $P_{c\gamma}(c_r;\gamma) > 0$ for $c_r > 0$, and $P_c(c_r;\gamma) \to \infty$ as $c_r \to \infty$.

Strict convexity captures rising marginal legal, electoral, administrative, and reputational costs of authorizing greater expected harm per participant. The cross-partial gives γ a transparent institutional interpretation among democracies: a larger γ raises the marginal political price of every positive c_r.

The state keeps $B_s > 0$ when the status quo survives; its policy payoff after protest-induced change is normalized to zero. With the proper planning distribution $\tau \sim U[0,1]$, the success probability and expected protest mass in the small-noise limit both equal $\hat{\tau}(c_r)$. Expected utility is

$$u_S(c_r) = (1 - \hat{\tau}(c_r))\, B_s - V(\hat{\tau}(c_r), c_r; w) - P(c_r;\gamma). \tag{3}$$

For $c_r \in (0, d_p)$, the marginal payoff is[6]

$$u_c(c_r) = \frac{B_s + V_N}{d_p} - V_c - P_c. \tag{4}$$

The term $(B_s + V_N)/d_p$ values the status quo and planned operational resources saved when deterrence reduces expected protest mass. The term V_c is the direct marginal operational cost of implementing harsher tactics. The term P_c is the marginal political cost of expected participant harm.

5. Thus $P_c \equiv \partial P(c_r;\gamma)/\partial c_r$, $P_{cc} \equiv \partial^2 P(c_r;\gamma)/\partial c_r^2$, and $P_{c\gamma} \equiv \partial^2 P(c_r;\gamma)/(\partial c_r \partial \gamma)$. The function's other argument remains fixed in each derivative.

6. Here $u_c \equiv \partial u_S/\partial c_r$ and $u_{cc} \equiv \partial^2 u_S/\partial c_r^2$. These derivatives vary the state's policy c_r and allow the induced expected mass $\bar{N}(c_r) = \hat{\tau}(c_r)$ to adjust, while fixing B_s, d_p, w, γ, and the functional forms of V and P.

Assumption 5 (Interior curvature)**.** The objective is twice continuously differentiable and strictly concave on $(0, d_p)$, with $u_{cc}(c_r) < 0$ and right derivative $u_c(0^+) > 0$.

Using equation (4), the condition $u_{cc} < 0$ is equivalent to

$$\frac{V_{NN}}{d_p^2} - \frac{2V_{Nc}}{d_p} + V_{cc} + P_{cc} > 0.$$

This condition allows planned operational cost to be nonlinear in expected protest mass and expected participant harm while keeping a unique policy response.

Proposition 2 (The optimal repression policy)**.** *Under Assumptions 2–5:*

(i) The state has a unique optimal policy on $[0, d_p]$*. The solution is interior and satisfies* $u_c(c_r^*) = 0$ *when the left derivative* $u_c(d_p^-) < 0$*; otherwise the state chooses the full-deterrence corner* $c_r^* = d_p$*.*

(ii) Expected harm above d_p *is dominated because the success threshold and expected protest mass are already clamped at zero while* V *and* P *are weakly increasing in* c_r*.*

Online Appendix A.5 (p. A-8) gives the proof.

3.5 Comparative statics.

The general technology separates effects that follow directly from the model from the condition governing the crowd-cost mechanism.

Proposition 3 (Comparative statics)**.** *At the interior solution:*[7]

(i) The policy comparative statics are

$$\frac{\partial c_r^*}{\partial B_s} = -\frac{1/d_p}{u_{cc}} > 0, \qquad \frac{\partial c_r^*}{\partial \gamma} = \frac{P_{c\gamma}}{u_{cc}} < 0,$$

$$\frac{\partial c_r^*}{\partial d_p} = \frac{B_s + V_N + c_r^* V_{Nc} - (c_r^*/d_p) V_{NN}}{d_p^2 u_{cc}}, \quad \frac{\partial c_r^*}{\partial B_p} = 0,$$

Thus $\partial c_r^*/\partial d_p < 0$ *when* $B_s + V_N + c_r^* V_{Nc} > (c_r^*/d_p) V_{NN}$.

(ii) For the crowd-cost shifter,

$$\frac{\partial c_r^*}{\partial w} = -\frac{V_{Nw}/d_p - V_{cw}}{u_{cc}}. \tag{5}$$

Hence a higher marginal cost of preparing for protesters increases expected participant harm if and only if

$$\frac{V_{Nw}}{d_p} > V_{cw}. \qquad \text{(Crowd-cost condition)}$$

(iii) A rise in B_s *lowers the equilibrium success probability, whereas a rise in* γ *raises it. When the crowd-cost condition holds, a rise in* w *lowers the success probability. A rise in* d_p *raises it whenever the stated condition for* $\partial c_r^*/\partial d_p < 0$ *holds.*

Online Appendix A.6 (p. A-9) gives the proof. The crowd-cost condition is the paper's central technological result. The term V_{Nw}/d_p is the additional planning cost saved when deterrence reduces expected protest mass. The term V_{cw} captures how the same shifter changes the direct marginal operational cost of harsher tactics. The first force raises the

7. Each expression $\partial c_r^*/\partial z$ or $\partial \hat{\tau}^*/\partial z$ is an equilibrium comparative-static derivative. It allows the optimizing policy, expected protest mass, and success threshold to adjust to primitive z while the other listed primitives and both cost-function families remain fixed. All derivatives of V on the right-hand side are evaluated at $(\hat{\tau}^*, c_r^*; w)$; derivatives of P are evaluated at $(c_r^*; \gamma)$.

state's optimal expected harm per participant; the second tempers that increase and can reverse it.

Two organizational technologies give the condition substantive content. Interpret w as the price of personnel and logistical inputs, and let $A(c_r)$ collect operational costs that do not vary with w. In a *deployment-separable* system,

$$V^D(\bar{N}, c_r; w) = w\bar{N} + A(c_r). \tag{6}$$

The input price raises the marginal cost of preparing for an additional protester, $V^D_{Nw} = 1$, without changing the direct marginal cost of the directive, $V^D_{cw} = 0$. Thus $\partial c_r^*/\partial w > 0$: higher personnel, transport, or crowd-control deployment costs make deterrence more valuable.[8]

In a *tactic-intensive* system, the same inputs also implement the authorized level of expected harm:

$$V^T(\bar{N}, c_r; w) = w(\bar{N} + \lambda c_r) + A(c_r), \qquad \lambda \geq 0. \tag{7}$$

Now $V^T_{Nw} = 1$ and $V^T_{cw} = \lambda$. The price increase raises expected participant harm when $\lambda < 1/d_p$ and lowers it when $\lambda > 1/d_p$, as Figure 2 shows.[9] The second case describes organizations in which specialized officers, less-lethal munitions, medical support, detention processing, or documentation requirements make stronger directives especially input-intensive. Equations (6) and (7) yield a discriminating implication: comparable

8. The derivative $\partial c_r^*/\partial w$ allows the optimal directive and anticipated protest mass to adjust to the input price w; B_s, d_p, γ, and the functional forms of A and P remain fixed.

9. The loading λ converts one unit of expected participant harm into the same operational-input units used to measure $\bar{N}$; its units are therefore units of anticipated participation per unit of harm. Because d_p and c_r share payoff units, $1/d_p$ has the same units as λ. The numerical cutoff is tied to those normalizations and is not directly portable across differently scaled data.

input-price increases lead the state to authorize more expected participant harm where they load mainly on personnel deployment and less expected participant harm where they load heavily on implementing force.

The institutional comparison follows directly from $P_{c\gamma} > 0$. Environments with stronger legal liability, judicial oversight, electoral accountability, or media scrutiny raise the marginal political price of expected harm and therefore reduce c_r^*. This gives the democratic setting a formal role through the cost schedule faced by the elected executive.

For the transparent benchmark, use

$$V(\bar{N}, c_r; w) = w\bar{N}, \qquad P(c_r; \gamma) = \gamma c_r^2.$$

This is the deployment-separable technology with $A(c_r) = 0$. Then $V_{Nw} = 1$, $V_{cw} = 0$, the crowd-cost condition holds strictly, and the solution reduces to

$$c_r^* = \min\left\{\frac{B_s + w}{2\gamma d_p}, d_p\right\}, \qquad \hat{\tau}^* = \max\left\{1 - \frac{B_s + w}{2\gamma d_p^2}, 0\right\}.$$

Figure 1 uses this additive-quadratic specialization to show the mechanism.

In the benchmark, a higher w raises expected participant harm. It increases the planned resources saved by reducing expected protest mass without changing the direct marginal operational cost of harsher tactics. A higher γ lowers c_r by steepening the marginal political-cost schedule. Greater intrinsic motivation d_p lowers the interior policy because motivated citizens are harder to deter. At the full-deterrence corner, c_r tracks d_p one for one. Online Appendix A.1 (p. A-2) maps the policy regions of this benchmark and plots its comparative statics in B_s and d_p.

The benchmark is the case $\lambda = 0$, in which a higher input price leaves the direct cost of harsher tactics unchanged and therefore always raises expected harm. Figure 2 adds the technology in equation (7) with $A(c_r) = 0$ and the same political cost $P(c_r; \gamma) = \gamma c_r^2$. Solving

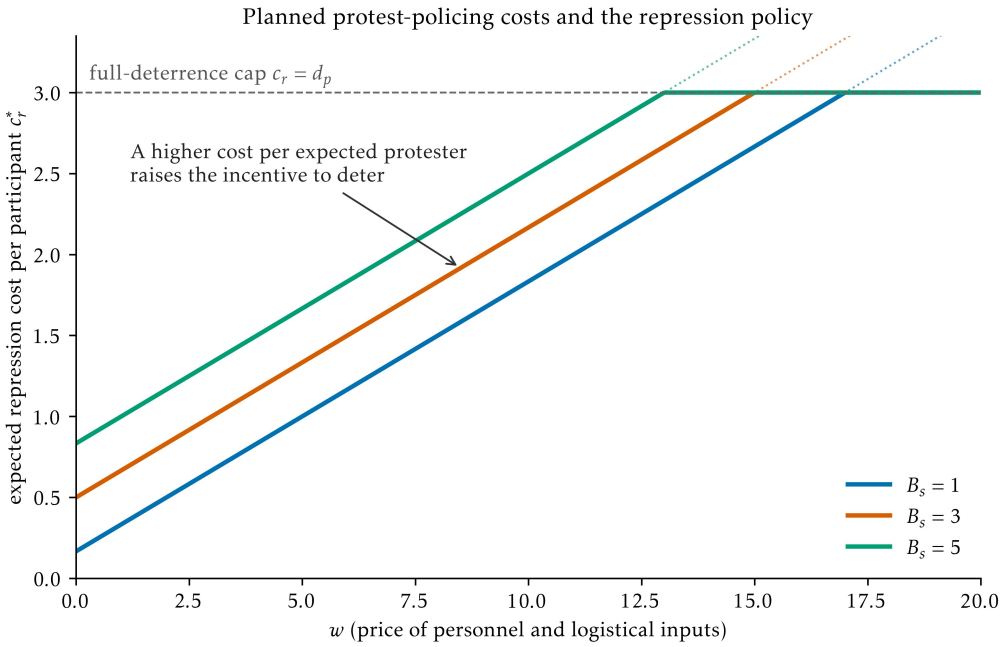


Figure 1: The crowd-management mechanism in the additive-quadratic benchmark, $V(\bar{N}, c_r; w) = w\bar{N}$ and $P(c_r; \gamma) = \gamma c_r^2$, with $\gamma = 1$ and $d_p = 3$. Solid lines show the effective policy for three values of the government's status quo stake; dotted lines show the unconstrained interior solution. A higher marginal cost of preparing for protesters raises expected participant harm until the full-deterrence cap binds.

the state's problem then gives the interior policy and its response to the input price,

$$c_r^* = \frac{B_s + w(1 - \lambda d_p)}{2\gamma d_p}, \qquad \frac{\partial c_r^*}{\partial w} = \frac{1 - \lambda d_p}{2\gamma d_p}.$$

Panel A fixes $B_s = 5$ and $d_p = 3$ and raises the input price w under four loadings λ. All four lines start at $B_s/(2\gamma d_p)$, and the line for $\lambda = 0$ reproduces the $B_s = 5$ line of Figure 1. The same price increase raises expected participant harm when $\lambda < 1/d_p$, leaves it unchanged when $\lambda = 1/d_p$, and lowers it when $\lambda > 1/d_p$. Panel B maps the sign of that response over d_p and λ. The boundary between its two regions depends on neither B_s nor γ, since B_s enters only the level of the policy and γ scales its response to w without changing its sign.

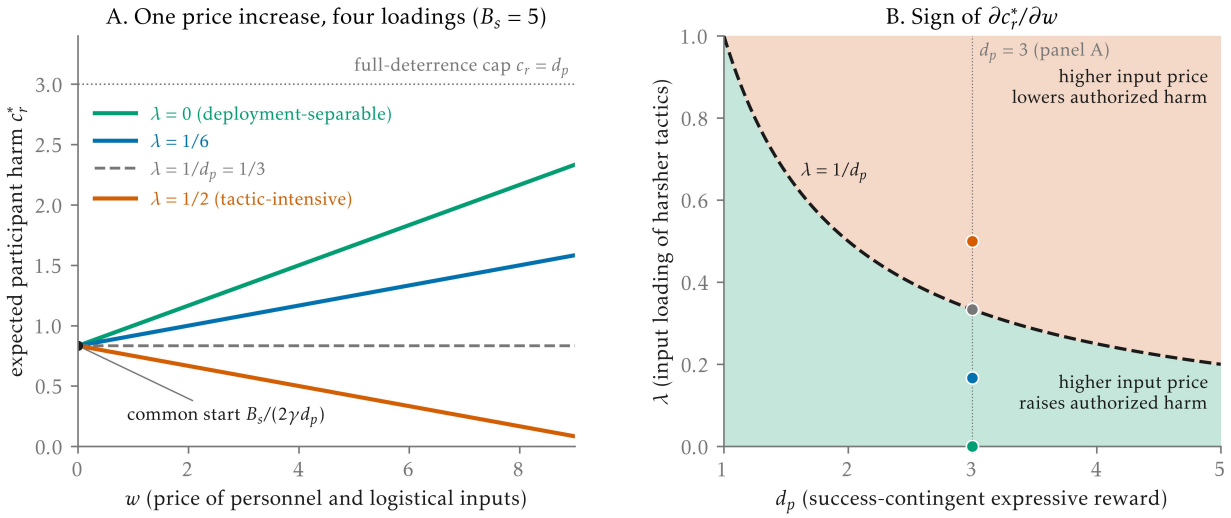


Figure 2: Reversal of the budget effect when the same inputs implement force. Both panels use the technology $V^T(\bar{N}, c_r; w) = w(\bar{N} + \lambda c_r)$ and the political cost $P(c_r; \gamma) = \gamma c_r^2$, with $\gamma = 1$. Panel A fixes $B_s = 5$ and $d_p = 3$ and varies the input price w for four loadings λ, and every policy shown is interior. Panel B shows the sign of $\partial c_r^*/\partial w$ over d_p and λ, with the boundary $\lambda = 1/d_p$ dashed and the four loadings of panel A marked at $d_p = 3$.

Two features of these results deserve emphasis.

First, the status quo prize B_s has an unambiguously positive effect at every interior optimum. The monotone cutoff equilibrium makes the deterrence technology the linear relation (2), so the marginal value of expected participant harm rises with the government's stake in the disputed policy.

Second, the direct partial effect of B_p equals zero under the benchmark's nonpivotality and separability assumptions. Policy stakes affect repression through the participation reward d_p and the government's status quo value B_s. The relevant distinction is between the prize's direct public-good component and the motivational and governmental channels through which politically important disputes alter behavior.

3.6 Directive costs and exposure-dependent political costs.

Accountability can operate at two moments. Some consequences follow the authorization itself: published rules of engagement, legal review, and press scrutiny of a directive impose costs whether or not anyone is hurt. Others follow what happens on the street: lawsuits, casualty investigations, and voters' reactions to documented injuries grow with the number of people actually harmed. The second kind depends on turnout as well as tactics. A harsher directive hurts each participant more but keeps some citizens at home, so the total number harmed first rises and then falls, reaching zero when no one protests. Accountability based on consequences therefore restrains moderate directives most, while a directive harsh enough to deter nearly everyone faces only the cost of authorizing it. This subsection adds that second cost and shows that the crowd-cost condition still determines how budget pressure moves the directive.

Formally, the baseline $P(c_r;\gamma)$ is the cost of authorization. In the small-noise limit, total expected harm in the population is

$$T(c_r) = c_r \hat{\tau}(c_r).$$

This aggregation uses the benchmark's homogeneous expected harm per participant and sharp small-noise participation.[10] Let $L(T;\eta)$ be a twice continuously differentiable,

10. At positive noise with homogeneous exposure, the corresponding object is $c_r \mathbb{E}_\tau[N^*(\tau)]$. With heterogeneous exposure c_{ri}, it is $\mathbb{E}_\tau[\int_0^1 n_i(\tau)c_{ri}\,di]$. Casualty, injury,

increasing, convex political cost of exposure, normalized by $L(0;\eta) = 0$, in addition to the directive cost P. The state then maximizes

$$u_S^E(c_r) = (1 - \hat{\tau})B_s - V(\hat{\tau}, c_r; w) - P(c_r; \gamma) - L(c_r\hat{\tau}; \eta).$$

At an interior policy its marginal payoff is[11]

$$u_c^E(c_r) = \frac{B_s + V_N}{d_p} - V_c - P_c - L_T\left(\hat{\tau} - \frac{c_r}{d_p}\right). \tag{8}$$

Exposure-dependent accountability changes the optimal level because total expected harm first rises and then falls as deterrence reduces the crowd. Differentiating equation (8) in the crowd-cost shifter w again gives $V_{Nw}/d_p - V_{cw}$ when the extended objective remains strictly concave, satisfies $u_{cc}^E < 0$ at the interior optimum, and L does not depend on w.[12] The crowd-cost condition therefore continues to determine the sign of $\partial c_r^*/\partial w$. The directive cost remains positive at full deterrence, while the exposure cost reaches zero with expected turnout.

3.7 An extension: Repression, protests, and emotions.

The benchmark holds success-contingent motivation fixed. To isolate one feedback, let citizens observe the directive c_r before choosing whether to participate and let $d_p(c_r) = d_0 + e(c_r)$. The observable object can be a public rule, announced tactic, or credible

or detention totals therefore map to T only after accounting for turnout and the distribution of exposure among participants.

11. Here $L_T \equiv \partial L(T;\eta)/\partial T$, evaluated at $T = c_r\hat{\tau}(c_r)$. The derivative varies c_r, allows $\hat{\tau}(c_r)$ and total expected harm to adjust, and fixes B_s, d_p, w, γ, η, and the cost-function families. Since $\partial\hat{\tau}/\partial c_r = -1/d_p$, $\partial(c_r\hat{\tau})/\partial c_r = \hat{\tau} - c_r/d_p$.

12. The required curvature is $u_{cc}^E = u_{cc} - L_{TT}(1 - 2c_r/d_p)^2 + 2L_T/d_p < 0$, where L_T and L_{TT} are evaluated at $T = c_r\hat{\tau}(c_r)$. Convexity of L alone does not ensure this inequality.

expectation formed from prior police practice. The threshold becomes

$$\hat{\tau}(c_r) = \max\left\{1 - \frac{c_r}{d_0 + e(c_r)}, 0\right\}.$$

At an interior threshold, differentiation gives[13]

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{d_0 + e(c_r) - c_r\, e'(c_r)}{(d_0 + e(c_r))^2}.$$

Define the elasticity of success-contingent motivation to expected participant harm as $\varepsilon(c_r) = c_r e'(c_r)/d_p(c_r)$.

Proposition 4 (Deterrence and backlash)**.** *At any policy satisfying* $0 < \hat{\tau}(c_r) < 1$*, with positive differentiable motivation* $d_p(c_r)$*:*[14]

(i) Expected participant harm deters, $\partial\hat{\tau}/\partial c_r < 0$*, if and only if* $\varepsilon(c_r) < 1$*.*

(ii) It backfires, $\partial\hat{\tau}/\partial c_r > 0$*, if and only if* $\varepsilon(c_r) > 1$*.*

Online Appendix A.7 (p. A-10) gives the proof. The extension converts the punishment puzzle into a scale-free condition. Anticipated repression deters when motivation rises less than proportionally with expected harm and recruits when it rises more than proportionally.

The emotional response also changes the directive the state is willing to choose. Index the strength of that response by $\alpha \geq 0$ and write $d_p(c_r; \alpha) = d_0 + \alpha e(c_r)$, so $\alpha = 1$ recovers the preceding specification. At an interior threshold, define the marginal deterrent return

$$D(c_r, \alpha) \equiv -\frac{\partial \hat{\tau}(c_r; \alpha)}{\partial c_r} = \frac{d_p(c_r; \alpha) - c_r \alpha e'(c_r)}{d_p(c_r; \alpha)^2} = \frac{1 - \varepsilon(c_r; \alpha)}{d_p(c_r; \alpha)}, \tag{9}$$

13. The derivative $\partial\hat{\tau}/\partial c_r$ varies the observable policy c_r and fixes d_0 and the response function $e(\cdot)$; $e'(c_r)$ is the ordinary derivative of that function evaluated at c_r.

14. Where the threshold is locally clamped at zero, its derivative is zero; at a transition into or out of this region, one-sided derivatives apply.

where $\varepsilon(c_r;\alpha) = c_r \alpha e'(c_r)/d_p(c_r;\alpha)$.[15]

Proposition 5 (Emotions and the state's policy)**.** *Suppose $d_0 > 0$, e is twice continuously differentiable, $e(c_r) \geq 0$, and $e'(c_r) \geq 0$. Let $c_r^*(\alpha) > 0$ be an interior optimum with $\hat{\tau}(c_r^*;\alpha) \in (0,1)$, and assume the state's extended objective is strictly concave and satisfies $u_{cc}(c_r^*;\alpha) < 0$. Then:*

(i) Any such optimum lies in the deterrence region: $\varepsilon(c_r^*;\alpha) < 1$.

(ii) Under the deployment-separable technology $V^D(\bar{N}, c_r; w) = w\bar{N} + A(c_r)$, a stronger nonzero emotional response reduces the state's chosen expected harm:

$$\frac{\partial c_r^*}{\partial \alpha} < 0 \qquad \text{if} \qquad e(c_r^*) + c_r^* e'(c_r^*) > 0.$$

Online Appendix A.8 (p. A-10) gives the proof. A state does not choose a locally backfiring directive at an interior optimum: such a directive increases expected protest while also carrying operational and political costs. Even before backfire occurs, stronger emotional responsiveness weakens the marginal deterrent return. In a deployment-separable system, the state responds by selecting less expected harm. Evidence that mobilization follows a coercive tactic is informative about this mechanism when the tactic causally changes both expected participant harm and success-contingent motivation. The timing concerns an observable ex ante policy. Across repeated episodes, prior realized violence can determine citizens' current expectations of that policy.

15. The partial derivative $\partial\hat{\tau}(c_r;\alpha)/\partial c_r$ varies the directive while fixing d_0, α, and the response function $e(\cdot)$. In the proof, $D_\alpha \equiv \partial D/\partial\alpha$, $G_\alpha \equiv \partial G/\partial\alpha$, and $G_{c_r} \equiv \partial G/\partial c_r$ hold the other displayed argument fixed. The derivative $\partial c_r^*/\partial\alpha$ varies the strength of the emotional response while allowing the state to reoptimize; B_s, w, γ, $A(\cdot)$, $P(\cdot)$, and $e(\cdot)$ remain fixed. At $\alpha = 0$, this parameter derivative is a right derivative.

4 Empirical implications and illustrative evidence.

The theory distinguishes the operational cost of preparing for the expected protest volume from the political cost of authorizing expected harm per participant. These costs affect the state's choice differently. A higher marginal cost of preparing for protesters makes deterrence more valuable, whereas the marginal legal, electoral, administrative, and reputational costs of the directive restrain the policy used to deter. The derived crowd-cost condition determines the direction of the crowd-cost effect when the same cost environment also changes the direct marginal operational cost of harsher tactics. Expected participant harm rises only when the planning resources saved through deterrence exceed that additional marginal operational cost. Equation (8) adds political consequences from total expected harm while preserving this sign condition.

A substantial literature connects fiscal consolidation to protest. A separate literature explains protest-policing tactics, mainly through threat perception, security diffusion, and legal change. Noakes and Gillham (2007) and Gillham (2011) attribute the turn toward more preemptive tactics to that perceived threat. Selmini and Di Ronco (2023, p. 206) describe the resulting tactics: no-protest zones, wider use of less-lethal weapons, arrests made before protests, and renewed surveillance of movements seen as problematic. None of these three accounts examines how the structure of police costs shapes the choice of tactics. The model's central prediction remains untested. The remainder of this section reads existing evidence through the model.

The model assigns separate roles to three political quantities. The government's B_s is the value of preserving the disputed status quo. Citizens' d_p is the participation-contingent reward from contributing to a successful movement. The public policy benefit B_p has no direct pivotality effect in the continuum benchmark, but politically important disputes can affect both d_p and B_s. Klein and Regan's (2018) cross-national protest-event analysis provides suggestive evidence about status quo stakes. They find that high concession costs, meaning demands that are expensive for the ruling coalition to accept, are associated with

coercive state responses. This pattern is consistent with the positive effect of B_s. They also find that high disruption costs encourage accommodation. That result points in the opposite direction from the additive crowd-cost benchmark, but it concerns economic and public-order losses tied to a protest's location, duration, and size. The model's w instead changes the marginal operational cost of preparing for an additional participant. The contrast shows why the two costs should not be combined into a single measure of how expensive a protest is for the government.

The observable counterpart of c_r has two parts: reach q_r, the probability that a participant is targeted, and conditional severity s_r, the harm a targeted participant suffers, from detention and injury to sexual violence, trauma, and death. Event records can code both, giving $c_r = q_r s_r$. Because the executive picks a directive from a fixed menu (Assumption 1), the two move together, and measuring both shows how a police organization packages them. Castro (2025) shows that this coding is feasible. She codes civilian complaints from Chile's 2019 protest cycle by tactic, municipality, and day along two dimensions close to ours, severity and selectivity, and validates them against media reports and police ammunition records. Her finding that selective high-severity tactics and low-severity tactics both increased protest is consistent with the backfire region of Proposition 4, though it does not estimate the model's motivational elasticity.

Estimating the operational cost $V(\bar{N}, c_r; w)$ requires the government's anticipated protest volume and the resources committed before an episode. The relevant inputs include scheduled personnel, overtime capacity, transport, equipment and consumables, temporary command infrastructure, detention capacity, and medical support. Benchmarking their prices against comparable labor and procurement costs measures opportunity cost across jurisdictions. Budget documents, deployment plans, procurement records, and after-action accounts can distinguish standing capacity from resources added in anticipation of protest. They can also estimate the loading λ in equation (7): input-price changes that mainly raise deployment cost predict greater expected participant harm,

while price changes that strongly raise the cost of implementing force predict restraint. Within-jurisdiction changes are especially informative because ordinary crime also affects resource demand and police capacity.

Zurich's 1980–81 protest cycle is the most explicitly documented case of the substitution the model describes. Wisler and Kriesi (1998, p. 99) report that the cycle required uniformed officers almost daily and forced overtime, and that no federal force could supply reinforcements. The police responded with small mobile units, the so-called TW-taxi groups, which cut a large operation from about four hundred officers to rarely more than two hundred. "In order to compensate for the reduced numbers of police and the lost dissuasive effect of large-scale units of riot police, the TW-taxi groups were granted wider latitude in the use of tear gas and rubber bullets" (p. 99), and a 1984 city directive codified that latitude (p. 100). The authors also attribute the change to demonstrators' dispersed tactics and a strong law-and-order coalition, so the case does not isolate the personnel margin. It does show a police organization substituting authorized force for deployed personnel, the response equation (6) predicts when the marginal cost of deployment rises.

In Lithuania, Juska and Woolfson (2012) record police allocations falling by 7.8 percent in 2009, with a further cut proposed at 21.3 percent. The authors report that real wages for rank-and-file officers fell by almost a fifth. The same period brought centralized command and tactics that, in the authors' account, relied on the threat of violence rather than on consent. Centralization let the authorities redirect a shrinking police budget toward managing potential unrest in the capital. Their explanation differs from ours. They attribute it to doubts about whether officers who had themselves lost pay under austerity would remain reliable, a channel distinct from deployment cost.

Officers in two further cases attribute violence to scarcity at the point of deployment. Greek riot police interviewed by Vasilaki (2016, p. 20) name numerical shortfall directly: "Operations are designed in a way that makes us violent. We are few against many." British officers interviewed after the 2011 riots attributed their loss of control to a lack

of resources, and the Metropolitan Police afterward sought water cannon and authorized baton rounds (O'Sullivan 2017, pp. 35–36). Inspectors in England and Wales record commanders reporting that "the powers are sufficient; it is the ability to implement them that is the challenge due to lack of resources" (HMICFRS 2021, p. 97). The same inspection found that the police lacked the capacity to make mass arrests in that scenario (p. 97). These accounts record officers and commanders reasoning about severity under a resource constraint, which is the reasoning the model formalizes.

Fiscal pressure also produced restraint in these same episodes, which the crowd-cost condition anticipates. Vasilaki (2016) reports that Greek riot-squad numbers rose during austerity. O'Sullivan (2017, p. 198) finds the Metropolitan Police extending protest liaison partly because it could reduce the police resources mobilized at protests. That extension lowered severity under the same budget pressure that produced the baton-round authorization. Whether a cost shock raises or lowers expected participant harm depends on how it loads across deployment and tactical implementation. The crowd-cost condition specifies which sign holds.

In Spain, police recruitment fell sharply after 2011, and enforcement shifted toward fines. Oliver Olmo and Urda Lozano (2015, p. 1322) name this pattern *burorrepresión*, a form of coercion and intimidation that stays out of public view. Authorities adopt it because visible violence against demonstrators generates public sympathy and solidarity with them (p. 1322). Oliver and Urda (2019, p. 84) add that many people cannot pay such fines during a crisis, so a fixed nominal fine imposes greater expected cost as incomes fall. Administrative fines raise c_r while consuming little operational capacity, and they also generate revenue. In the notation of equation (7), such an instrument has a loading λ near zero, and the revenue it raises offsets part of its operational cost. The Spanish literature attributes the shift to the political cost of visible force, which the model represents through the exposure cost $L(T;\eta)$ and the directive cost $P(c_r;\gamma)$. The model identifies a second margin operating at the same time. This case does not show how to separate the two, and

that remains an empirical question.

Institutions shift $P(c_r;\gamma)$ by changing the consequences of authorizing a directive for elected officials. Published rules of engagement, judicial preclearance, legislative oversight, internal compliance review, and press access expose the authorization itself to scrutiny (della Porta and Reiter 1998, pp. 10–11). Civil liability, casualty investigations, and electoral reactions to documented injuries instead load on $L(T;\eta)$. Aytaç, Schiumerini, and Stokes (2017) study backlash movements in Turkey, Brazil, and Ukraine and find that governments with stable electoral support were more willing to use harsh measures, whereas electorally insecure governments had stronger incentives to reduce violence. This pattern is consistent with the model's claim that a lower marginal electoral cost permits greater expected participant harm. At a broader level, Davenport and Armstrong (2004) find that democracy reduces human-rights violations only above a threshold of democratic development. Their country-level outcome is not specific to protest directives, but the nonlinear result is consistent with interpreting γ as meaningful institutional variation among democracies. Research designs can use the timing of these institutions to distinguish changes in the marginal directive cost P_c from changes in the marginal exposure cost L_T.

Existing evidence also points to the motivational channel, while showing that participants need not respond uniformly. Adam-Troian, Çelebi, and Mahfud's (2020) cross-sectional study of 523 Yellow Vest protesters finds that exposure to police violence is positively associated with intentions to attend future demonstrations and to sacrifice for the movement. Perceived loss of significance and identification with the Yellow Vests mediate those associations. C. M. Sullivan and Davenport's (2017) panel study of the Republic of New Africa finds organizational backlash but different individual responses: members who personally experienced repression became more likely to participate, while members who were not directly repressed withdrew. These findings are consistent with a positive motivational response for some citizens, but they also caution against treating

$e(c_r)$ as homogeneous.

Event records can first describe the joint variation in anticipated turnout, planned operational inputs, coercive reach, conditional harm, and institutional constraints. Suitable designs can then use institutional reforms and predetermined changes in labor or procurement costs to isolate particular margins. Two existing studies already report associations of this kind. H. Sullivan (2021, p. 587) finds that Mexican subnational governments with bureaucratic capacity are less likely to respond to protest events with repression. The crowd-cost condition would explain that association through the marginal cost of managing participants. The same study qualifies the accountability channel: electoral competition does not significantly reduce the likelihood of a repressive response (p. 587). Beck (2023) finds that United States municipalities funding one percent more of their budgets from fines and fees experienced four percent more police killings. Beck leaves the mechanism open (p. 175), and the model suggests one candidate: where fiscal strain makes deploying officers more expensive, harsher tactics become a cheaper way to manage the population police must control. The extension is only suggestive, because routine street policing lacks the coordination problem through which a protest crowd responds to the directive. The comparative statics specify the expected pattern. Status quo stakes raise expected participant harm, and stronger marginal accountability lowers it. The crowd-cost shifter w raises expected harm only where the resources saved through deterrence exceed its effect on the marginal operational cost identified by the crowd-cost condition.

5 Conclusion.

This paper explains why a government that finds large protests expensive to manage may threaten harsher repression before people mobilize. The logic has two steps. Expected harm discourages some citizens from joining. A smaller expected crowd lets the government plan fewer officers, vehicles, detention spaces, and supplies. Harsher repression

becomes attractive when the resources saved by preparing for a smaller crowd exceed the added operational and political costs of authorizing force. What matters is the cost of managing one more protester. The total price of maintaining a police force is a different quantity.

This tradeoff also explains why governments facing similar protests can respond differently. Consider an economic downturn that cuts tax revenue while raising demand for unemployment benefits, emergency transfers, and other social spending. The resulting fiscal pressure leaves less money for police overtime, recruitment, and equipment. If officials still expect a large protest, they may be unable to expand deployment as they otherwise would. Where revising the use-of-force directive is comparatively cheap, the executive may compensate by authorizing tactics that expose each participant to greater expected harm, hoping that stronger deterrence allows a smaller force to manage the episode. The result still depends on how the police organization produces force. If stronger tactics require specialized officers, equipment, documentation, medical support, or detention capacity, the budget pressure can instead produce restraint. Legal review, political oversight, and electoral accountability further change the price that an elected government pays when it authorizes expected harm. Repression depends on which resources have become scarce and on how the directive uses them.

Citizens complete the argument because the crowd is not fixed in advance. Each person weighs the expected harm of participation against the satisfaction of helping a protest succeed. A stronger directive usually discourages participation and makes policy change less likely. Anger can reverse that result, but only when the added motivation grows faster than the expected harm. Repression can therefore anger citizens without producing a larger protest; anger and backfire are separate outcomes. When the state anticipates a stronger emotional response, it chooses less expected harm in the deployment-separable benchmark.

The argument points to a dynamic question for future research. Governments announce

directives and plan deployments before an episode, but commanders and citizens also learn after turnout and injuries become visible. Comparing the initial directive with later tactical revisions would show when governments use an early threat to avoid a costly crowd, when operational demands force restraint, and when observed violence creates the mobilization that the directive was meant to prevent.

References

Adam-Troian, Jais, Elif Çelebi, and Yara Mahfud. 2020. "'Return of the Repressed': Exposure to Police Violence Increases Protest and Self-Sacrifice Intentions for the Yellow Vests." *Group Processes & Intergroup Relations* 23 (8): 1171–86. https://doi.org/10.1177/1368430220920707.

Amenta, Edwin, Neal Caren, Elizabeth Chiarello, and Yang Su. 2010. "The Political Consequences of Social Movements." *Annual Review of Sociology* 36:287–307. https://doi.org/10.1146/annurev-soc-070308-120029.

Arias, Enrique Desmond, and Daniel M. Goldstein, eds. 2010. *Violent Democracies in Latin America.* Durham, NC: Duke University Press. https://doi.org/10.1215/9780822392033.

Aytaç, S. Erdem, Luis Schiumerini, and Susan C. Stokes. 2017. "Protests and Repression in New Democracies." *Perspectives on Politics* 15 (1): 62–82. https://doi.org/10.1017/S1537592716004138.

Aytaç, S. Erdem, and Susan C. Stokes. 2019. *Why Bother? Rethinking Participation in Elections and Protests.* Cambridge: Cambridge University Press.

Battaglini, Marco. 2017. "Public Protests and Policy Making." *The Quarterly Journal of Economics* 132 (1): 485–549. https://doi.org/10.1093/qje/qjw039.

Beck, Brenden. 2023. “Police Killings and Municipal Reliance on Fine-and-Fee Revenue.” *RSF: The Russell Sage Foundation Journal of the Social Sciences* 9 (2): 161–81. https://doi.org/10.7758/RSF.2023.9.2.07.

Belur, Jyoti. 2010. “Police and the Use of Deadly Force.” In *Permission to Shoot? Police Use of Deadly Force in Democracies*, edited by Jyoti Belur, 1–20. New York: Springer. https://doi.org/10.1007/978-1-4419-0975-6_1.

Bor, Jacob, Atheendar S. Venkataramani, David R. Williams, and Alexander C. Tsai. 2018. “Police Killings and Their Spillover Effects on the Mental Health of Black Americans: A Population-Based, Quasi-Experimental Study.” *The Lancet* 392 (10144): 302–10. https://doi.org/10.1016/S0140-6736(18)31130-9.

Bueno de Mesquita, Ethan. 2010. “Regime Change and Revolutionary Entrepreneurs.” *American Political Science Review* 104 (3): 446–66. https://doi.org/10.1017/S0003055410000274.

Bueno de Mesquita, Ethan, and Mehdi Shadmehr. 2023. “Rebel Motivations and Repression.” *American Political Science Review* 117 (2): 734–50. https://doi.org/10.1017/S0003055422000600.

Carey, Sabine C. 2006. “The Dynamic Relationship between Protest and Repression.” *Political Research Quarterly* 59 (1): 1–11. https://doi.org/10.1177/106591290605900101.

Carlsson, Hans, and Eric van Damme. 1993. “Global Games and Equilibrium Selection.” *Econometrica* 61 (5): 989–1018. https://doi.org/10.2307/2951491.

Castro, Francisca. 2025. “Deterrence or Defiance? How Severity and Selectivity Shape Protest Responses to Repression.” *Mobilization: An International Quarterly* 30 (4): 387–408. https://doi.org/10.17813/1086-671X-30-4-387.

Çelebi, Elif, Jais Adam-Troian, and Yara Mahfud. 2022. "Positive Links between Exposure to Police Violence, PTSD, and Depression Symptoms among Yellow Vests Protesters in France." *Journal of Interpersonal Violence* 37 (5–6): 2267–88. https://doi.org/10.1177/0886260520935863.

Chau, Tak-Huen, Mai Hassan, and Andrew T. Little. 2025. "Communication, Coordination, and Surveillance in the Shadow of Repression." *American Journal of Political Science* 69 (3): 995–1009. https://doi.org/10.1111/ajps.12904.

Chenoweth, Erica, Evan Perkoski, and Sooyeon Kang. 2017. "State Repression and Nonviolent Resistance." *Journal of Conflict Resolution* 61 (9): 1950–69. https://doi.org/10.1177/0022002717721390.

Cooper, Hannah, Lisa Moore, Sofia Gruskin, and Nancy Krieger. 2004. "Characterizing Perceived Police Violence: Implications for Public Health." *American Journal of Public Health* 94 (7): 1109–18. https://doi.org/10.2105/AJPH.94.7.1109.

Curtice, Travis B., and Brandon Behlendorf. 2021. "Street-Level Repression: Protest, Policing, and Dissent in Uganda." *Journal of Conflict Resolution* 65 (1): 166–94. https://doi.org/10.1177/0022002720939304.

Davenport, Christian. 2007a. "State Repression and Political Order." *Annual Review of Political Science* 10:1–23. https://doi.org/10.1146/annurev.polisci.10.101405.143216.

———. 2007b. *State Repression and the Domestic Democratic Peace.* Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511510021.

Davenport, Christian, and David A. Armstrong. 2004. "Democracy and the Violation of Human Rights: A Statistical Analysis from 1976 to 1996." *American Journal of Political Science* 48 (3): 538–54. https://doi.org/10.1111/j.0092-5853.2004.00086.x.

De Jaegher, Kris, and Britta Hoyer. 2019. "Preemptive Repression: Deterrence, Backfiring, Iron Fists, and Velvet Gloves." *Journal of Conflict Resolution* 63 (2): 502–27. https://doi.org/10.1177/0022002717750450.

della Porta, Donatella. 1995. *Social Movements and the State: Thoughts on the Policing of Protest.* EUI Working Paper RSC 95/13. Florence: European University Institute, Robert Schuman Centre. https://cadmus.eui.eu/handle/1814/1388.

della Porta, Donatella, and Herbert Reiter. 1998. "Introduction: The Policing of Protest in Western Democracies." In *Policing Protest: The Control of Mass Demonstrations in Western Democracies*, edited by Donatella della Porta and Herbert Reiter, 1–32. Minneapolis: University of Minnesota Press. https://doi.org/10.5749/j.ctttv1tv.3.

deMeritt, Jacqueline H. R. 2016. "The Strategic Use of State Repression and Political Violence." In *Oxford Research Encyclopedia of Politics.* Oxford University Press. https://doi.org/10.1093/acrefore/9780190228637.013.32.

DeVylder, Jordan, Lisa Fedina, and Bruce Link. 2020. "Impact of Police Violence on Mental Health: A Theoretical Framework." *American Journal of Public Health* 110 (11): 1704–10. https://doi.org/10.2105/AJPH.2020.305874.

DeVylder, Jordan E., et al. 2017. "Prevalence, Demographic Variation and Psychological Correlates of Exposure to Police Victimisation in Four US Cities." *Epidemiology and Psychiatric Sciences* 26 (5): 466–77. https://doi.org/10.1017/S2045796016000810.

Gillham, Patrick F. 2011. "Securitizing America: Strategic Incapacitation and the Policing of Protest since the 11 September 2001 Terrorist Attacks." *Sociology Compass* 5 (7): 636–52. https://doi.org/10.1111/j.1751-9020.2011.00394.x.

HMICFRS (Her Majesty's Inspectorate of Constabulary and Fire & Rescue Services). 2021. *Getting the Balance Right? An Inspection of How Effectively the Police Deal with Protests.* London: HMICFRS. https://hmicfrs.justiceinspectorates.gov.uk/publications/getting-the-balance-right-an-inspection-of-how-effectively-the-police-deal-with-protests.

Juska, Arunas, and Charles Woolfson. 2012. "Policing Political Protest in Lithuania." *Crime, Law and Social Change* 57:403–24. https://doi.org/10.1007/s10611-012-9363-4.

Klein, Graig R., and Patrick M. Regan. 2018. "Dynamics of Political Protests." *International Organization* 72 (2): 485–521. https://doi.org/10.1017/S0020818318000061.

Little, Andrew T. 2016. "Communication Technology and Protest." *The Journal of Politics* 78 (1): 152–66. https://doi.org/10.1086/683187.

Machado, Fabiana, Carlos Scartascini, and Mariano Tommasi. 2011. "Political Institutions and Street Protests in Latin America." *Journal of Conflict Resolution* 55 (3): 340–65. https://doi.org/10.1177/0022002711400864.

Morris, Stephen, and Hyun Song Shin. 1998. "Unique Equilibrium in a Model of Self-Fulfilling Currency Attacks." *American Economic Review* 88 (3): 587–97.

———. 2003. "Global Games: Theory and Applications." In *Advances in Economics and Econometrics: Theory and Applications, Eighth World Congress*, edited by Mathias Dewatripont, Lars Peter Hansen, and Stephen J. Turnovsky, 56–114. Cambridge: Cambridge University Press. https://doi.org/10.1017/CCOL0521818729.003.

Nam, Taehyun. 2007. "Rough Days in Democracies: Comparing Protests in Democracies." *European Journal of Political Research* 46 (1): 97–120. https://doi.org/10.1111/j.1475-6765.2006.00645.x.

Noakes, John, and Patrick F. Gillham. 2007. "Police and Protester Innovation since Seattle." *Mobilization: An International Quarterly* 12 (4): 335–40. https://doi.org/10.17813/maiq.12.4.hk88jk1mw3036302.

Nordås, Ragnhild, and Christian Davenport. 2013. "Fight the Youth: Youth Bulges and State Repression." *American Journal of Political Science* 57 (4): 926–40. https://doi.org/10.1111/ajps.12025.

O'Sullivan, Aidan. 2017. "Policing Protest in an Age of Austerity: How the Metropolitan Police Service (MPS) Responded to Anti-Austerity Movements after the Financial Crash." PhD diss., University of Liverpool. https://doi.org/10.17638/03010425.

Oliver, Pedro, and Jesús-Carlos Urda. 2019. "The Repression of Protest in Spain after 15-M: The Development of the Gag Law." *Social Justice* 46 (2–3): 75–99.

Oliver Olmo, Pedro, and Jesús-Carlos Urda Lozano. 2015. "Bureau-Repression: Administrative Sanction and Social Control in Modern Spain." *Oñati Socio-Legal Series* 5 (5): 1309–28.

Pierskalla, Jan Henryk. 2010. "Protest, Deterrence, and Escalation: The Strategic Calculus of Government Repression." *Journal of Conflict Resolution* 54 (1): 117–45. https://doi.org/10.1177/0022002709352462.

Reny, Tyler T., and Benjamin J. Newman. 2021. "The Opinion-Mobilizing Effect of Social Protest against Police Violence: Evidence from the 2020 George Floyd Protests." *American Political Science Review* 115 (4): 1499–1507. https://doi.org/10.1017/S0003055421000460.

Ritter, Emily Hencken. 2014. "Policy Disputes, Political Survival, and the Onset and Severity of State Repression." *Journal of Conflict Resolution* 58 (1): 143–68. https://doi.org/10.1177/0022002712468724.

Selmini, Rossella, and Anna Di Ronco. 2023. "The Criminalization of Dissent and Protest." *Crime and Justice* 52:197–231. https://doi.org/10.1086/727553.

Su, Yen-Pin. 2015. "Anti-Government Protests in Democracies: A Test of Institutional Explanations." *Comparative Politics* 47 (2): 149–67. https://doi.org/10.5129/001041515814224444.

Sullivan, Christopher M. 2016. "Undermining Resistance: Mobilization, Repression, and the Enforcement of Political Order." *Journal of Conflict Resolution* 60 (7): 1163–90. https://doi.org/10.1177/0022002714567951.

Sullivan, Christopher M., and Christian Davenport. 2017. "The Rebel Alliance Strikes Back: Understanding the Politics of Backlash Mobilization." *Mobilization* 22 (1): 39–56. https://doi.org/10.17813/1086-671X-22-1-39.

Sullivan, Heather. 2021. "Subnational State Capacity and the Repression of Protest: Evidence from Mexico." *Political Research Quarterly* 74 (3): 587–98. https://doi.org/10.1177/1065912920919494.

Vasilaki, Rosa. 2016. *Policing the Crisis in Greece: The Others' Side of the Story*. GreeSE Paper 98. London: Hellenic Observatory, London School of Economics. https://www.lse.ac.uk/Hellenic-Observatory/Assets/Documents/Publications/GreeSE-Papers/GreeSE-No98.pdf.

Wisler, Dominique, and Hanspeter Kriesi. 1998. "Public Order, Protest Cycles, and Political Process: Two Swiss Cities Compared." In *Policing Protest: The Control of Mass Demonstrations in Western Democracies*, edited by Donatella della Porta and Herbert Reiter, 91–116. Minneapolis: University of Minnesota Press. https://doi.org/10.5749/j.ctttv1tv.7.

Wood, Elisabeth Jean. 2003. *Insurgent Collective Action and Civil War in El Salvador*. Cambridge: Cambridge University Press. https://doi.org/10.1017/CBO9780511808685.

Worden, Robert E. 1995. "The 'Causes' of Police Brutality: Theory and Evidence on Police Use of Force." In *And Justice for All: Understanding and Controlling Police Abuse of Force*, edited by William A. Geller and Hans Toch, 31–60. Washington, DC: Police Executive Research Forum.

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The use of AI tools in preparing and checking this manuscript is disclosed here.[16]

16. The author wrote the initial manuscript, including its formal proofs. Claude Opus 5.5 (Anthropic) was used to review and lightly polish portions of the text. Astra 6 (OpenAI Codex) was used to probe the formal results for counterexamples through Python-based numerical searches across alternative functional forms and broad parameter ranges. The author reviewed all AI-assisted changes, verified the proofs and computational results, and takes full responsibility for the manuscript.